SO, THE WHOLE IDEA IS TO EXTEND THE LIFETIME OF THE SENSOR AND BATTERY OPTIMIZATION AND AN INTELLIGENT ROUTING PROTOCOL. IRRESPECTIVE OF THE APPLICATION AND TYPE OF SENSOR. AND THIS IS FOR THE MULTI SINK APPLICATIONS.

MODULE - 1

So first the clusters are formed, and CH is elected based on the residual energy and the nodes sense the data and sends data packets to the CH then there a DQN algorithm is applied at the CH only because of their less number. Then the CH filters out the unwanted data. How it does this is there will be 2 queues present in the CH namely normal priority queue and high priority queue, where the normal data will be sent through normal priority queue where it checks the whether the received data is above the threshold or not, if the data is not above the threshold then that data can be discarded (like if the temp sensor is saying the temp is 22c for every second which we do not need) and if the data sensed is above the threshold then it will be sent into the normal priority queue and in that queue the data packet waits for its turn to come to be sent to the base station, if it takes more than Tx seconds to be sent to the base station then we can consider that data is expired and can be discarded which we can call it as a buffer. And as for the High priority queue, it is like a bypass queue which the immediately sends the data if there is a sudden spike in the sensed data even if that is below the threshold (like say if the threshold is 50 and the sensed data has spiked from 5 to 48 suddenly), these kind of data should be immediately sent to the base station from there to the database and from there the client can access. We can define the queue length and if the queue size got exceeded, we may drop the data that stayed more than or equal to Tx seconds.

And after a fixed number of rounds the CH role is re-evaluated and a new CH may be selected based on the voting system. And the DQN is applied on the newly selected CH. Which should be optimal. AND WE SHOULD NOT FORGET THE CORE IDEA IS TO EXTEND THE LIFE OF A SENSOR, BATTERY OPTIMIZATION AND INTELLIGENT ROUTING PROTOCOL.

And I want to know where it is better to aggregate the data either at CH or base station. And also, how to mathematically select the CH and how we must define the threshold and I want the threshold to be dynamic over time.

So until here we have seen how the data is sensed and sent to the cluster.

MODULE – 2

Now from the CH how the data is sent to the base station (sink). Since there are multiple sinks which are stationary. We are not strictly following a single pattern to send the data from the CH to base station. And we can do bulk transmission of data which is energy efficient.

Here what I'm planning is to calculate

- ➔ Distance to each sink
- ➔ Estimated Delay
- ➔ Queue length
- ➔ Link quality
- ➔ Residual energy of CH
- ➔ Recent Success/ Failure

So here we are not fixing that a particular CH should only send data to a particular sink.

Instead, we are making an intelligent routing protocol where the CH calculates how far the nearest sink is and if that sink is busy then it should try to send the data to another sink or CH which is less busy. And it should be aware of its delay.

And I am using NS-3 in ubuntu, pytorch, C++ environment, and a DQN agent.

And I also want to know what mathematical tests like Wilcoxon method and Friedman test should be done to prove our work. And need the mathematical formulae to calculate every detail in the idea.

And Can this idea be simulated and is worth trying?